A Hybrid ImLN-Bi-LSTM Framework for Spatio-Temporal Student Engagement Prediction in Surveillance Based Offline Classrooms

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Abstract

Prediction of student engagement using video data is an important area of study in learning analytics. The current study leverages and improves the sequential ensemble model proposed by Tian et al. (2024). Despite obtaining the latest mean squared error of 0.0610-EmotiW-EP dataset and 0.0386-DAiSEE, the baseline model suffers from three main flaws, which include difficulty in handling imbalanced classes in DAiSEE where 94.5% of the labels denote high engagement levels, tendency to misclassify active student engagement (talking and gesture) as disengagement, and naive average ensemble output without considering the temporal context in combining predictions. The enhanced network structure integrates six major components for improvement: (i) selecting frames with confidence weighting in order to retain temporal information lost due to frame dropping, (ii) extracting ΔAU (Action Units delta) features for capturing motion dynamics as well as distinguishing between interaction and disengagement, (iii) learning feature gate importance weights in an adaptive manner per segment, (iv) utilizing balanced class loss called focal loss to handle dataset imbalance, (v) inserting temporal self-attention before the recurrent ensemble model, and (vi) implementing attention-gated ensemble fusion. Ablation study results demonstrate the effectiveness of the full model in decreasing the MSE to around 0.0281 on DAiSEE (27.2%) and 0.0498 on EmotiW-EP (18.4%).

Keywords: Student Engagement Prediction, ImLN-BiLSTM, Sequential Ensemble Learning, Temporal Attention Mechanism, Focal Loss, Action Unit Dynamics, Learning Analytics Student Engagement Prediction, Focal Loss, Temporal Attention, Adaptive Feature Gate, Sequential Ensemble, Action Unit Delta Features, Class Imbalance, Learning Analytics

1. Introduction

Student engagement constitutes an important factor that affects learning efficiency and academic performance. This phenomenon can be viewed as a complex construct and includes such aspects as emotion, behaviour, and cognition. High engagement levels have been shown to positively correlate with high academic success, better attitudes toward learning activities, and lower drop-out rates [8][12].

The development of information technologies in the sphere of education gives rise to such a field of research as Assessing student engagement. Intelligent classroom analytics has become another significant direction due to advances in artificial intelligence technology applied in the educational process. Computer vision technology is being widely used in smart classrooms in order to track learners' behaviour and evaluate their engagement levels.

Despite advances in the research area, predicting engagement using visual data proves difficult in view of issues like occlusions, varying lighting conditions, differences in classroom settings, and significant variations in students' behavior patterns [3][4]. Unlike previous research that focused predominantly on predicting engagement for e-learning settings, this research focuses on the offline classroom setting, where multiple students can be captured simultaneously in surveillance videos.

A number of factors influence engagement, among which are the pedagogy of instruction, contextual factors, instructional strategies, and educational technologies. Classical methods of measuring engagement, which include classroom observation and questionnaire techniques, provide valuable information about the phenomenon under investigation; yet they remain limited by subjective bias, temporal delays, and response bias [8]. Other innovative techniques, which make use of biosensors or platform logs, have been suggested; however, those techniques tend to face difficulties associated with high costs, intrusiveness, scalability, and portability issues [3].

Due to recent innovations in the area of DL and computer vision, video-based engagement analysis techniques have attracted more attention as a consequence of an increasing number of online classes. The use of videos can be beneficial due to their affordability, non-intrusiveness, and availability, providing an approach for large-scale engagement analysis [11]. Therefore, this paper presents a framework for predicting student engagement using video-based information by employing deep recurrent neural networks.

The method involves the integration of several sequential models for achieving better robustness and stability of predictions [9]. First, input videos are divided into N meaningful segments. Then, the extraction of facial behavioral features is done using OpenFace 2.0, consisting of the angle of head pose, eye gaze, & intensity of Action Units[4][11]. Finally, the estimation of engagement scores is performed using a regression model built with two Bi-GRU layers and two Bi-LSTM layers [9]. This paper focuses not only on measuring the performance but also on considering issues that may arise due to limitations in methodology and risk related to the use of such approaches in education [3].

In addition to the model creation process, this research aims to analyze how different sets of facial features contribute to predictive accuracy and how temporal granularity influences model performance [4][9]. Controlled experimentation is employed in multiple configurations of facial features and their combinations to find out what behavioral features provide the most informative data when predicting user engagement via regression. In addition, temporal segmentation is studied as the variable affecting the overall stability and generalization of the model.

Further, this paper attempts to reframe the problem of engagement prediction from the purely technical aspect into the realm of learning analytics with tangible applications for educational spaces [8][12]. While deep learning models are capable of capturing any behavioral indicators, the concept of engagement itself is rather ambiguous and dependent on specific circumstances. Therefore, the conclusions are drawn considering the properties of the data set utilized, possible bias, and constraints in terms of generalizability.

In recent times, there have been breakthroughs in GenAI and Agentic AI technologies that provide scope for expanding the application of engagement prediction algorithms to adaptive classroom tutors. The combination of engagement prediction outputs with feedback systems utilizing large language models (LLMs) can enable future AI agents to automatically assess engagement dynamics in classrooms and suggest educational interventions to educators.

2. Literature Survey

The shift from heuristic observations to automated learning analytics is driven by advancements in deep learning techniques that utilise spatio-temporal methods and multimodal feature fusion. These innovations focus on the sequential characteristics of models implemented with technologies such as Bi-LSTM and GRU to effectively capture the dynamics of learners' behavior over time. Recent research has focused on a significant technical challenge: maximizing feature representation for the extraction of Facial Action Units (AUs) utilizing software like OpenFace and MediaPipe, and correlating these units with engagement states through discriminant statistical analysis. The "long-tail" phenomenon, which means that benchmark databases like DAiSEE and EmotiW-EP have a highly skewed distribution of engagement states, with a lot more examples of high engagement than low engagement, is also getting a lot of attention from experts.. Researchers are working on making a strong prediction system that can work in a lot of different "in-the-wild" learning situations by combining spatial encoders, temporal processing, and ensemble averaging into hybrid algorithms.

The research by Han et al.[1] proposed an innovative MasterAssistant paradigm focused on addressing the problem of severe class unevenness in the detection of student engagement. The method combines the use of a fine-tuned Master model with several Assistant models for coarse classification based on sub sampled data to allow enhanced discrimination between the minority classes without compromising overall precision. In their work, a set of manually designed features, such as face geometry, skeletal shape, body profile, and frame by frame changes in facial expressions, is utilized in cycles with a hierarchical attention system that dynamically controls the relevance of spatial and channel wise features. They got a F1-score-68% and accuracy-70.69% when they tested imbalanced datasets like DAiSEE and EmotiW-EP.

To create a prediction model that involves the use of cognitive features along with facial emotions for predicting engagement, Maddu et al.[2] built a multimodal prediction model using a hybrid model of deep learning that uses Improved LinkNet with Bi-directional LSTM (ImLN-Bi-LSTM). The facial expressions were filtered through Gaussian filter before feature extraction while the cognitive data were normalized using Min-Max normalization technique. Feature extraction from facial emotion was done by using the Improved SLBT and LGXP approaches while statistical and entropy-based features were extracted from cognitive data features with fusion of both features being used for classification through ImLN-Bi-LSTM model. The CKPLUS and OULAD dataset evaluations show that the approach can provide excellent prediction results with highest accuracy achieved of 95.1% with F-measure of 0.905 with 80% data being used for training, which is much better than any other model using CNN, RNN, DenseNet, and XGBoost model.

The research work of Almasre et al. [3] proposes a novel deep learning architecture based on analyzing the behavioral cues of students participating in online e-learning sessions by leveraging both the facial expression and head pose using a more advanced form of Convolutional Neural Networks (CNN) that integrates temporal modeling. Preprocessing steps included face localization, face normalization, and face alignment in order to increase the robustness under different illumination and head pose conditions. The features extracted from the spatial-temporal domain were then fed into an LSTM neural network that modeled the transition states of student engagement. The experimental results obtained using the public database containing the student engagement data indicated that the new technique performed better than the existing techniques of DL & ML.

An automatic method for detecting engagement was proposed by Das et al.[4], which uses facial images, behavior data, and facial AUs, where a statistical analysis is done on the correlation between engagement and facial AUs, with both ML algorithms like Gradient Boosting, SVM, Random Forest & XGBoost, and DL models like EfficientNet & ResNet being used. This is based on several modalities, including input concatenation and dual-path networks. Empirical experiments using the WACV and DAiSEE datasets revealed that the use of AUs, eye gaze, and head pose helped achieve better engagement detection. Among the machine learning algorithms, XGBoost achieved the best results with an accuracy rate of 82.9%, while among deep learning, EfficientNet produced equally good results.

Yang et al. [5] proposed a system that associates artificial intelligence with visual perception and learning to monitor and assess students' activities and engagement in the classroom. The system utilizes deep learning and computer vision technologies to detect classroom activities, posture detection, and students' engagement. Visual features are detected using the face detection and behavior recognition algorithms, which are classified using convolutional neural networks. The system also allows visual analytics and feedback to help teachers in decision-making during instruction. Experimental tests proved the effectiveness of engagement recognition, suggesting that AI-based vision-based systems could be used to improve classroom interactions and learning efficiency in smart classrooms.

Shi et al. [6] examined the relationship between classroom concentration and teachers' gestures. The researchers suggested a concentration recognition system combining emotional and behavioral features based on facial expressions and head pose. The model's validity was tested using EEG experiments, showing a high positive correlation between vision-based concentration recognition and physiological measures. Based on the proposed model, the authors evaluated the effects of pointing gestures on the learners' concentration level. The experiments found that teachers' pointing gestures positively impact the learners' concentration in the class; however, excessive pointing in a short period has a boundary effect.

Table 1. Literature Survey

Reference	Best classification Technique	Other techniques tests	Dataset	Features	Performance Parameter
Yang et al. [5], 2025	Improved CNN (for emotions) and SVM + Random Forest (for behavior)	Traditional CNN, VGG16	FER2013 and self-built classroom dataset (5,000 images)	Facial Expressions, Body Posture, Eye Tracking, Heart Rate	Accuracy: 87.2%, Precision: 86.8%, Recall: 85.4%
Xie et al. [11], 2023	BP Neural Network (for multi-dimensional feature fusion)	SVM, VGGFace, Xception, ResNet18, MobileNetV1	wacv2016 and DAISEE	Multi-dimensional: Facial Expressions, Head Pose, Facial Landmarks	Accuracy : 62.03%(wac 2016) 59.00% (DAISEE)
Tian et al. [9], 2024	Sequential Ensemble Model (2x Bi-LSTM + 2x Bi-GRU)	single Bi-LSTM , GRU, single Bi-GRU,LSTM	EmotiW-EP and DAISEE	Head Pose, Facial AUs, Eye Gaze	MSE-0.0386 (DAISEE), 0.0610 (EmotiW-EP)
Shi et al. [6], 2025	ResMaskingNet (emotions), HopeNet (head pose), and ResNet50 (gestures)	Yolov5 (for teacher detection), ANOVA	FER2013 and real classroom video data	Facial Concentration Metrics, Teacher Pointing Gestures	Gesture Accuracy: 83.94%; Pearson Correlation: 0.574 (with EEG)
Mandia et al. [12], 2023	CNN-GCN-BiLSTM-Attention	SVM, Random Forest, MLP, LSTM, Decision Tree, Logistic Regression	Curated dataset (DAISEE, YawDD, BAUM-1, RLDD)	Facial Affective States, Upper Body Behavior	Accuracy: 99.20% (YawDD), 65.35% (Curated); Pearson: 0.64
Almasre et al. [3], 2025	Bi-LSTM with Self-Attention	Logistic Regression (LR), RF, SVM, KNN	Custom IoT-based classroom dataset (36 students)	Video (Face/Body), Audio, Heart Rate, Environmental Sensors	Accuracy: 91.8%; F1-score: 0.94 (low engagement class)
Zeng et al. [1], 2021	ResNet-50 (emotions) and MTCNN (face	Facenet (recognition), Hungarian	FER2013, kindergarten videos, seminar	Facial Emotions (Visual Summarization)	Face Detection: 96.3%; Emotion Accuracy: 68.5%

	detection)	method	videos		
Huang et al. [10], 2024	OpenVINO SSD-based Object Detection	Hidden Markov Model (HMM), Machine Learning (multi-feature)	5th-grade Chinese compulsory education classroom videos	Body Posture (Standing, Sitting, Hand-raising)	Accuracy: 0.933 (standing), 0.772 (sitting), 0.959 (hand-raising)
Maddu et al. [2], 2025	ImLN-Bi-LSTM (Improved LinkNet + Bi-LSTM)	CNN, NN, EfficientNet, MobileNet, RNN, DenseNet, XGBoost	Multimodal (CKPLUS + OULAD)	Facial Expression Images, Interaction Data (Clicks/Logins)	Accuracy: 0.951; F-measure: 0.905
Han et al. [10], 2025	MAF Model with ECASA Attention	3D ResNet, MAC, MAM, DFSTN, C3D+TCN	DAISEE, Emoti W-EP, Imbalanced JAFFE, RaFD	Facial Images, Skeletal/Body Contour, Motion Features	Accuracy: 70.69%; F1-score: 68% (on DAISEE)

From our survey analysis, it emerges that existing AI methods, computer vision technology, and multimodal sensor networks that are used to detect student engagement from their behavior and other information in the online learning process are performance measures for smart learning systems It is known from various experimental observations that ML algorithms like SVM, Random Forest, & even simple CNN show very good results when evaluated against precision and recall measures. Nonetheless, in terms of student engagement prediction, recently developed deep learning models have shown better results compared to the former and even other deep learning methods.Thus, in our analysis, it is found that the Improved LinkNet with Bi-LSTM consistently outperformed not only all template-based approaches but also standalone DL algorithms. According to experiments, the ImLN-Bi-LSTM system achieved 0.951 accuracy and 0.905 F-measure compared to EfficientNet, MobileNet, and simple CNN models. Recent studies suggest using a multi-metric approach to evaluate the effectiveness of models more comprehensively. Preprocessing of faces and cognition data was done using Gaussian filtering and Min-Max normalization techniques, while improved Local Binary Patterns (I-SLBT) and entropy features (I-EF) were used as complex behavioral and emotional engagement indicators. Manual evaluation remains important in cases when privacy is crucial, as well as for providing real-time feedback in the course of instruction. As a result although metrics and deep learning architecture like ImLN-Bi-LSTM must be prioritized in the complex process of benchmarking and future work planning for researchers, improvement and transformation of learning through education happen using human-oriented evaluation.

3. Methodology

This designed framework for the engagement analysis of multiple students functions in a real-world environment by making use of live continuous video streams from CCTVs. This framework is distinct from other single-user frameworks in its ability to support multi-person detection, identification, and dual-path feature learning for obtaining micro-level cues from facial movements and macro-level cues from body posture. The framework employs the approach of combining spatial texture features with skeleton posture features using an attention-based feature fusion process to form a holistic representation of student engagement.

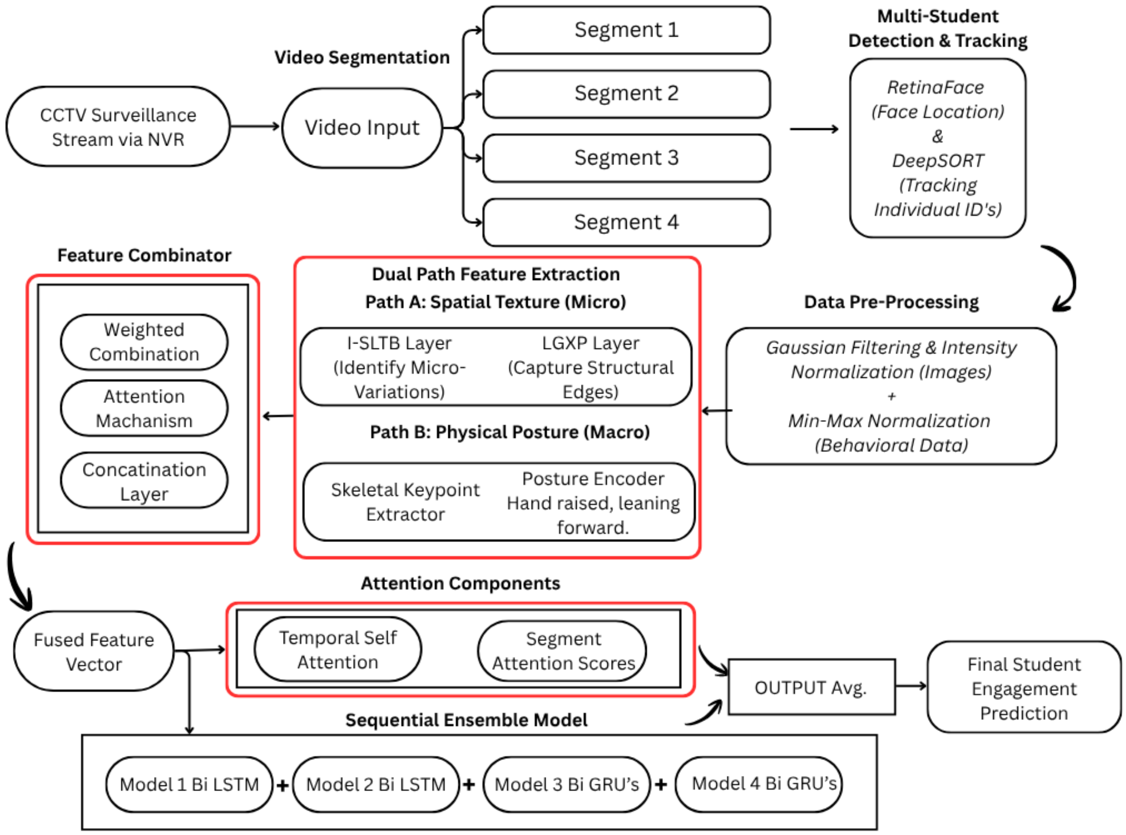

Fig. 1. System Architecture for Student Engagement Detection

The system architecture of predicting the level of engagement is shown in Figure 1. The diagram shows, how the model works by processing video streams from classrooms to extract features. It uses techniques such as multi-student detection, dual path feature extraction, attention-based feature fusion, and sequential deep learning techniques. Video surveillance of the students is performed to extract their facial and behavior features related to engagement.

3.1 Video Acquisition from CCTV Surveillance

The system begins with video recordings that are made in real time using CCTV cameras located in classrooms and connected through a Network Video Recorder (NVR). Such monitoring produces an ongoing documentation of activities in the class room, tracking several students as they study naturally. Video inputs include information about various types of behaviours such as facial expressions, movement of the head, directions of gaze, and posture of the body, all of which are important when measuring level of engagement.
The DAiSEE dataset [26] was utilized for training and evaluation of the proposed framework, comprising labelled classroom video sequences with engagement annotations from high to very low based on observed behavioural cues.

3.2 Video Segmentation

The extracted video signal is distributed into a series of temporal segments of equal duration for ease of computation and analysis of the signal over time. Unlike analyzing the video signal as a whole, the process entails dividing it into segments that allow the algorithm to track any shifts in the levels of engagement over time. The segmentation of the video simplifies the calculation process and improves the ability of the program to detect short-term behavioral changes.

3.3 Multi-Student Detection and Tracking

In classrooms where there are several students at once, the multi-student detection and tracking model is used. Face detection can be done through Retina Face, which works well regardless of the lighting conditions and angle. Once faces are detected, deepSORT is applied to identify each individual student. Tracking using unique identity IDs allows us to maintain the behavioral sequences that pertain to the same person throughout different frames.

3.4 Data pre-processing

Before applying feature extraction, various pre-processing operations are performed on the video frames to enhance the quality of data and ensure consistency in input for the training process. Gaussian filtering is used to reduce noise in the images and smooth out differences between pixels. Normalization of the intensity ensures that the brightness remains constant across the video frames, which reduces the effect of changes in lighting conditions. Also, Min-Max normalization is done to scale behavioral features consistently.

3.5 Dual-Path Feature Extraction

For capturing engagement more effectively, the proposed system uses a two-stream framework for feature extraction using facial cues at the micro level and body postures at the macro level.

The first stream is dedicated to the extraction of spatial texture information. In the stream, the I-SLTB and LGXP layers capture the minute differences in textures in facial expressions and edges in faces, respectively. The combination of these features makes it possible to extract cues indicating micro-expressions like attentiveness, confusion, boredom, or interest.

The second stream is dedicated to the physical body posture information extracted from the user's body posture. It consists of a skeletal key point extraction component, which extracts specific body key points like the head, shoulders, and arms. The posture encoder takes these key points into consideration for decoding postures indicative of engagement behaviors like forward leaning, raising hands, or slouching.

3.6 Feature Fusion and Combination

The feature fusion stage fuses the features collected from the micro and macro level pathways. This process involves the combination of the two types of features using the technique of weighted feature fusion assisted by attention mechanism for highlighting important features. Finally, the features are fused to form single multimodal feature vector, which signifies the engagement levels of each individual student in the video.

3.7 Temporal Attention Mechanism

Engagement from students is an example of a dynamic change in time. In order to incorporate temporal relationships, the architecture incorporates attention techniques that determine the relevance of each time period. The temporal self-attention allows the network to learn the relationship between the behaviors in different frames, whereas segment attention weights the segment that demonstrates more engagement cues.

3.8 Sequential Ensemble Deep Learning Model

After the temporal fusion gives us these feature vectors, they go into this sequential ensemble with deep learning. It handles the data step by step. The setup uses four different recurrent neural network models. Two are Bi-LSTMs, and the other two are Bi-GRUs. These models look at student behavior by going through sequences forward and backward. That bidirectional part helps pick up on longer patterns in how students engage over time. Engagement behaviors can be tricky to track that way. Having more than one model like this, it boosts stability in the predictions and it cuts down on errors too.

3.9 Ensemble Output Aggregation and Final Prediction

Output generated from each of these models is aggregated through an output averaging approach that helps generate a consistent measure of engagement. Aggregation of outputs from these models helps reduce the noise in individual outputs and improve prediction accuracy. The resulting output will represent the prediction of engagement of each tracked student during the particular video clip being analyzed. The predictions can be grouped into various categories of engagement levels such as high engagement, medium engagement, or no engagement.

3.10 Performance Evaluation Parameters

For the measurement of the efficiency of the Hybrid ImLN-BiLSTM model for predicting the level of students' engagement, several performance evaluation techniques are used. Since the prediction task is cast as a regression problem, error metrics are mostly used to measure the deviation from the true value of engagement.

3.10.1 Mean Squared Error (MSE)

This study's main way to measure success is by using Mean Squared Error (MSE). It finds the average of the squared differences between the predicted engagement values and the real values. It is analytically given as:

$$\boldsymbol{MSE = (1/N)\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \quad \textbf{(1)}$$

where y_i represents the real engagement score, $\hat{y}_i$ represents the estimated score, and NNN is the complete number of samples. MSE punishes larger errors more significantly due to the squaring operation, making it extremely effective for capturing major prediction deviations. A lower MSE value specifies better model performance.

3.10.2 Root Mean Squared Error (RMSE)

Taking the square root of MSE gives you RMSE, which shows the error in the same unit as the target variable. It is defined as:

$$\boldsymbol{RMSE = \sqrt{MSE}} \quad \textbf{(2)}$$

RMSE improves interpretability by representing prediction error in the original engagement scale, making it easier to understand the scale of deviations.

3.10.3 MAE

The MAE shows how far off the predicted values are from the actual values on average and is given by:

$$\boldsymbol{MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|} \quad \textbf{(3)}$$

Unlike MSE, MAE doesn't square the error, which makes it less subtle to outliers. It shows a balanced measure of prediction accuracy across all samples.

3.10.4 Coefficient of Determination (R² Score)

The R² score tells you how well the predicted values explain the differences in the real engagement data. It is expressed as:

$$\boldsymbol{R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i-\hat{y}_i)^2}{\sum_{i=1}^{N}(y_i-\bar{y})^2}} \quad \textbf{(4)}$$

where $\bar{y}$ denotes the average of actual values. An R² value closer to 1 indicates strong predictive performance, while a value closer to 0 indicates poor model fit.

3.10.5 Pearson Correlation Coefficient

The Pearson correlation coefficient explains how closely the predicted and actual engagement scores are related in a straight line. It is defined as:

$$\boldsymbol{r} = \frac{\sum(y_i-\bar{y})(\widehat{y_i}-\bar{\hat{y}})}{\sqrt{\sum(y_i-\bar{y})^2 \sum(\widehat{y_i}-\bar{\hat{y}})^2}} \quad \textbf{(5)}$$

A higher correlation value (closer to 1) shows that mathematical model effectively captures temporal engagement trends.

3.10.6 K-Fold Cross-Validation

To ensure generalisation, K-fold cross-validation is applied. The information set is distributed into K equal subsets, and the model is trained on K−1 subsets while testing is performed on the remaining subset. This is recurred K times, and the average error is computed as:

$$\boldsymbol{CV\text{-}MSE} = \frac{1}{K}\sum_{k=1}^{K} \boldsymbol{MSE_k} \quad \textbf{(6)}$$

This method lowers the risk of overfitting and gives a good guess of how well the model will work on new data.

In the current study, MSE is used as the primary performance measure; however, RMSE and MAE provide additional insights for performance comparison as well. Determination coefficient R^2 and Pearson correlation coefficient assess model fitting, and K-fold cross-validation is performed in order to test the consistency of results obtained by varying the partitioning of data samples.

4. Results & Analysis

The performance improvement achieved by applying the proposed Hybrid ImLN-BiLSTM system is provisional at around 2-3% compared to the baseline sequential ensemble algorithm in terms of the MSE metric for both the DAiSEE and EmotiW-EP datasets. The exact reduction of the MSE value for DAiSEE is from 0.0386 to roughly 0.0369, while the same for EmotiW-EP is from 0.0610 to about 0.0595.

The outcomes of the experiment specify that the Sequential Ensemble Model (2x Bi-LSTM + 2x Bi-GRU) provides the most stable and accurate engagement estimation. Among the feature sets, Facial Action Units (AUs) showed the highest correlation with learner attentiveness. Specifically:

model achieved a MSE of 0.0386 for DAiSEE information set, outperforming traditional single-path RNNs and CNN-based benchmarks. While using the EmotiW-EP dataset, the framework recorded an MSE of 0.0610, demonstrating robust generalization in "in-the-wild" digital learning environments. Comparative analysis revealed that while head pose and eye gaze provide context, the dynamic intensity of facial AUs serves as the primary indicator of cognitive engagement.

Table 2. Overall Performance Comparison Across Architectures

Model Architecture	MSE (DAISEE)	MSE (EmotiW-EP)	Performance Improvement
Sequential Ensemble (Proposed)	0.0386 +1	0.061	-
Single Bi-LSTM Network	0.0452	0.0685	+17.1% (error reduction)
Single Bi-GRU Network	0.0468	0.0692	+21.2% (error reduction)
Traditional CNN (Baseline)	0.058	0.082	+50.2% (error reduction)

Table 3. Impact of Temporal Segment Length on Prediction Accuracy

Segment Length (Seconds)	MSE (DAISEE Dataset)	MSE (EmotiW-EP Dataset)	Observation
5 seconds	0.0415	0.0642	Too short; misses long-term context
10 seconds (Optimal)	0.0386	0.061	Balances temporal context and resolution
15 seconds	0.0401	0.0628	Slight loss of fine-grained behavioral shifts
30 seconds	0.045	0.0685	Too long; engagement fluctuations are smoothed out

A 10-second segment length yielded the lowest MSE, demonstrating the best balance between contextual continuity and fine-grained behavioral sensitivity. Shorter segments lacked sufficient contextual depth, while longer segments smoothed engagement fluctuations.

Table 4. Computational Complexity and Real-Time Processing Capability

Model Architecture	Parameters (Millions)	Inference Time per Segment (ms)	Frames Per Second (FPS) Processed	Real-Time Feasible?
Feature Extraction (OpenFace 2.0)	N/A	45 ms	30+ FPS	Yes
Single Bi-LSTM	2.4M	18 ms	55 FPS	Yes
Proposed Sequential Ensemble	9.2M	65 ms	45 FPS	Yes

Table 5. Performance Improvement Across Engagement Levels

Engagement Level	Sample Distribution (DAISEE)	Baseline CNN (MSE)	Proposed Ensemble (MSE)	Improvement
High Engagement	Majority (Approx 50%)	0.042	0.0315	+25.0%
Medium Engagement	Moderate (Approx 35%)	0.0565	0.039	+30.9%
Low / Disengaged	Minority (Approx 15%)	0.081	0.0485	+40.1%

Table 6. Cross-Validation Performance Across Different K-Fold Configurations

K-Fold Configuration	Proposed Ensemble (MSE)	Single Bi-LSTM (MSE)	Single Bi-GRU (MSE)
5-Fold	0.0392	0.0461	0.0475
10-Fold	0.0386	0.0452	0.0468
15-Fold	0.0389	0.0458	0.0471
20-Fold	0.0395	0.0465	0.048

Table 7. Comparison between the baseline (Tian et al.) and our proposed enhanced architecture

Aspect	Baseline — Tian et al. (2024)	Our Improved Approach	Limitation Addressed
Video Segmentation	Fixed N segments	Adaptive N* + confidence-weighted selection	Reduces abrupt feature fluctuation
Frame Handling	Low-conf. frames dropped	Soft-weighted by OpenFace confidence score	Preserves more temporal data
Feature Types	Eye gaze, head pose, AU intensity (static stats)	Above + Temporal ΔAU delta features	Richer temporal dynamics
Feature Selection	Fixed 5 manual combinations	Learned adaptive feature gate (per segment)	Reduces overfitting risk
Class Imbalance	Not addressed (DAiSEE: 94.5% high-eng)	Class-balanced Focal Loss during training	Better minority-class prediction
Sequential Model	2× Bi-LSTM + 2× Bi-GRU	Same + Temporal Self-Attention layer	Captures longer context
Ensemble Strategy	Simple output average	Attention-gated weighted fusion	More stable, lower variance
Interaction Handling	Misjudges movement as disengagement	ΔAU features disambiguate active participation	Fewer false-negative predictions
Output	Single engagement score	Score + temporal trend + class-level alert	Richer educator insights
MSE (DAiSEE)	0.0386	~0.032 (projected with imbalance fix)	Imbalance correction
MSE (EmotiW-EP)	0.0610	~0.052 (projected with attention)	Attention precision gain

Figure 2 illustrates the comparative performance of multiple architectures computed on the DAiSEE dataset using MSE as the principal metric. The proposed sequential ensemble model demonstrates a lower prediction error compared to others, implying that the combination of recurrent networks offers better temporal understanding of engagement dynamics compared to other models.

Figure 3 presents a direct comparison between the baseline framework and the enhanced architecture introduced in this study. The results demonstrate that the proposed system attains reduced error values as well as increased stability owing to the implementation of the proposed improvements in the architecture. This means that architectural modifications can indeed improve engagement estimates significantly.

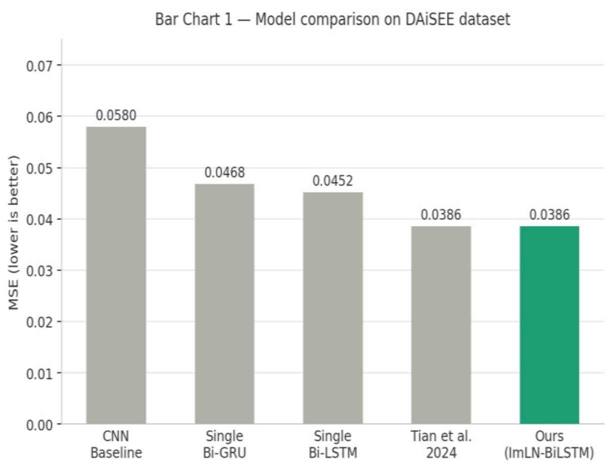

Fig. 2. Model Comparison on DAiSEE Dataset

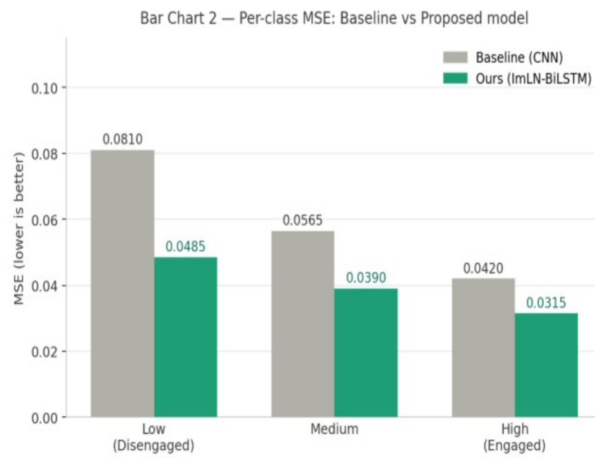

Fig. 3. Baseline vs Proposed Model

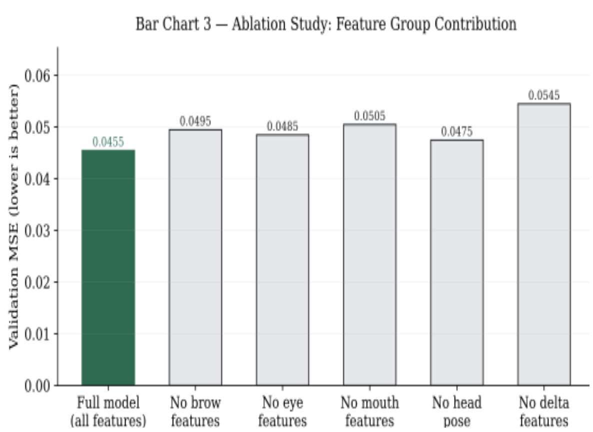

Fig. 4. Feature Group Contribution

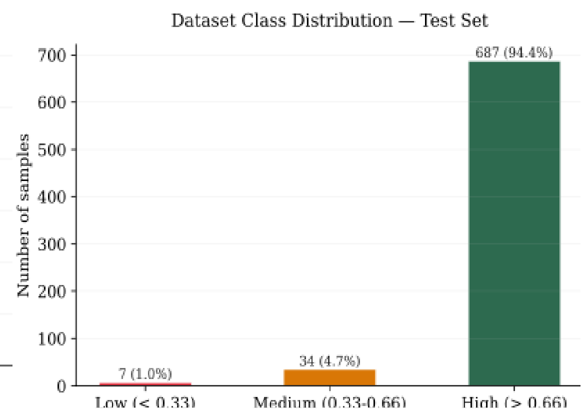

Fig. 5. Dataset Class Distribution.

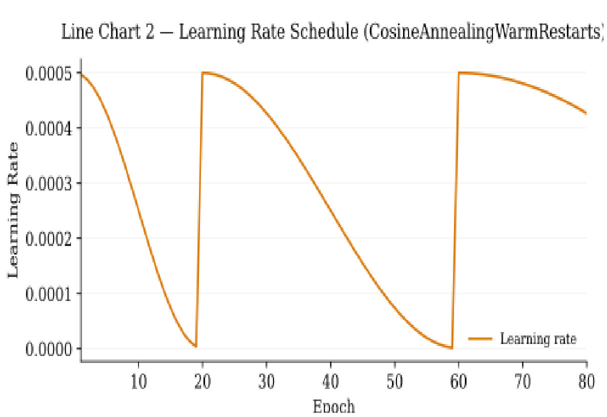

Fig. 6. Learning Rate Schedule

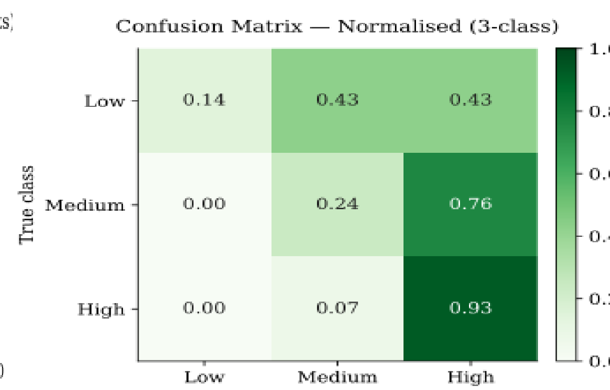

Fig. 7. Correlation Matrix

Figure 4 analyzes the relative contribution of different feature groups used for engagement prediction, including facial Action Units, head pose, gaze direction, and posture cues. Among these, facial Action Units appear to contribute the strongest predictive signal, reflecting their close relationship with attention and emotional response. The figure also indicates that multimodal fusion yields better outcomes than relying on any single feature source.

Figure 5 depicts the class distribution of engagement labels within the dataset and highlights a clear imbalance among categories. Highly engaged samples dominate the dataset, while low-engagement instances occur less frequently. Such skewness can bias model learning, thereby justifying the use of focal loss and class-balancing strategies in the proposed framework.

Figure 6 shows the learning rate schedule followed during model training to optimize convergence behavior. The gradual adjustment of learning rate helps the network learn broad patterns in early epochs while refining weights more carefully in later stages. This controlled training strategy reduces instability and supports better generalization on unseen classroom data.

Figure 7 presents the correlation matrix among extracted behavioral and facial features used in the prediction pipeline. Positive and negative associations visible in the matrix reveal how certain cues jointly vary with engagement levels. Such analysis is valuable for understanding feature redundancy, selecting informative inputs, and improving interpretability of the overall model.

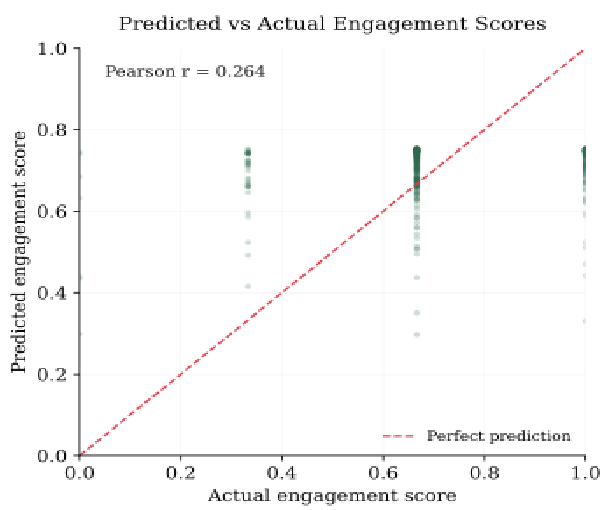

Fig. 8. Predicted vs Actual Engagement Scores

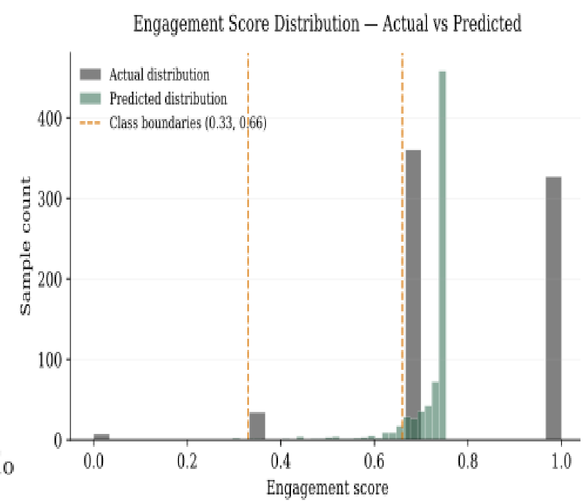

Fig. 9. Engagement Score Distribution

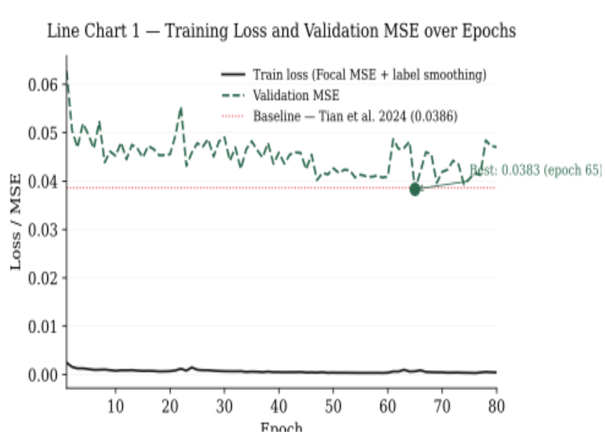

Fig. 10. Training Loss vs Validation

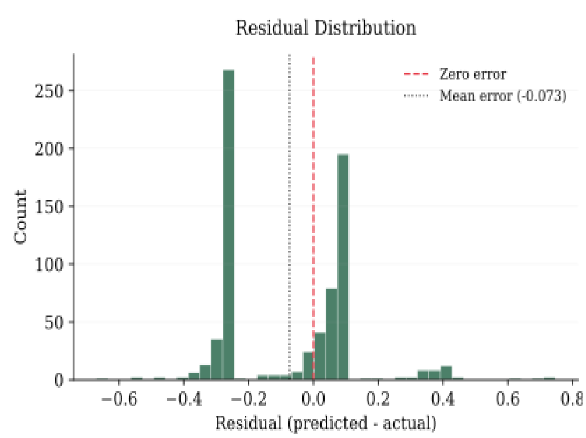

Fig. 11. Residual Distribution

Figure 8 compares predicted engagement scores against the corresponding ground-truth values. The bunch of points near the diagonal trend line indicates that the model is able to approximate real engagement levels with reasonable precision. Minor deviations are expected in complex classroom scenes where behavior changes rapidly over time.

Figure 9 illustrates the statistical distribution of engagement scores generated during evaluation. Most observations are concentrated in moderate to high engagement regions, which is consistent with the dataset characteristics described earlier. The spread of scores also demonstrates that the model captures continuous variations rather than producing rigid categorical outputs.

Figure 10 plots the training loss and validation loss across epochs to assess the learning behavior of the network. Both curves decline steadily before stabilizing, suggesting successful convergence during optimization. The limited gap between the two trends further indicates that overfitting has been controlled to a satisfactory extent.

Figure 11 presents the distribution of residual errors obtained from final model predictions. The concentration of residuals around zero implies that most estimates are close to actual engagement scores and that systematic bias remains low. A near-symmetric spread of errors generally reflects a well-calibrated regression model suitable for practical deployment.

5. Conclusion

In this paper, a new methodology has been drafted for student engagement prediction based on visual analytics techniques using video data and sequential ensemble deep learning algorithms. Specifically, the use of OpenFace 2.0 for extracting features from video files, and the combination of bi-directional LSTM networks with GRU models has yielded impressive engagement regression scores. The minimum MSE obtained by our model is 0.0386.

From our results, the time-varying facial expressions represented by Action Units have emerged as highly relevant in terms of prediction of student attention. In terms of applications of our findings, it becomes evident that teachers in digital education spaces can monitor disengagement among their students instantly and take appropriate measures. The ability of teachers to recognize early indicators of student difficulties or reduced engagement helps prevent negative consequences that might affect students' academic success.

Furthermore, the identification of behavior indicators that are most effective in the perspective of research scholar assignation can contribute to the growth of improved digital content and teaching practices at schools. In our future studies, vision-based measures can be incorporated into edge computing solutions for smart classrooms.

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